

Midterm Exam Equation Sheet

ISE 390, Fall 2026, The University of Alabama in Huntsville

1 Statistical Summaries

$$\bar{x} = \frac{1}{n} \sum^n x_i$$

$$s^2 = \frac{1}{n} \sum^n (x_i - \bar{x})^2$$

$$s^2 = \frac{1}{n-1} \sum^n (x_i - \bar{x})^2$$

$$s^2(\text{alt}) = \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right)$$

$$s = \sqrt{s^2}$$

2 Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\text{Permutations of } k \text{ from } n \text{ objects: } \frac{n!}{(n-k)!}$$

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$\text{Combinations of } k \text{ from } n \text{ objects: } \frac{n!}{k!(n-k)!}$$

3 Random Variables

$$\mu_x = \sum_x x p(x)$$

$$\mu_x = \int_{-\infty}^{\infty} x f(x) dx$$

$$\bar{X} = \frac{1}{n} X_1 + \dots + \frac{1}{n} X_n$$

$$\mu_{\bar{X}} = \mu$$

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}$$

$$\text{Cov}(X, Y) = \mu_{(X-\mu_X)(Y-\mu_Y)} = \mu_{XY} - \mu_X \mu_Y$$

$$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

$$\sigma_{c_1 X_1 + \dots + c_n X_n}^2 = c_1^2 \sigma_{X_1}^2 + \dots + c_n^2 \sigma_{X_n}^2 - 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n c_i c_j \text{Cov}(X_i, X_j)$$

$$\sigma_x^2 = \sum_{i=1}^n p_i (x_i - \mu_x)^2 = \sum_x (x^2 p_x) - \mu^2$$

$$\sigma_x^2 = \int_{-\infty}^{\infty} f(x) (x - \mu_x)^2$$

$$\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

$$F(x) = \int_{-\infty}^x f(t) dt$$

$$\mu_{h(X)} = \int_{-\infty}^{\infty} h(x) f(x) dx$$

$$\mu_{Y|X=x} = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy$$

$$f(x, y) = f_X(x) f_Y(y)$$

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

$$\mu_{c_1 X_1 + \dots + c_n X_n} = c_1 \mu_{X_1} + \dots + c_n \mu_{X_n}$$

$$\sigma_{c_1 X_1 + \dots + c_n X_n}^2 = c_1^2 \sigma_{X_1}^2 + \dots + c_n^2 \sigma_{X_n}^2$$

4 Measurement Error

$$c_{\text{best}} = \frac{\sigma_A^2}{\sigma_A^2 + \sigma_B^2} \text{ and } 1 - c_{\text{best}} = \frac{\sigma_B^2}{\sigma_A^2 + \sigma_B^2}$$

$$\sigma_{c_1 X_1 + \dots + c_n X_n} \leq |c_1| \sigma_{X_1} + \dots + |c_n| \sigma_{X_n}$$

$$\sigma_U \approx \left| \frac{dU}{dX} \right| \sigma_X$$

$$\frac{\sigma_A}{A} = \sigma_{\ln A}$$

$$\sigma_U \approx \sqrt{\left(\frac{\partial U}{\partial X_1} \right)^2 \sigma_{X_1}^2 + \left(\frac{\partial U}{\partial X_2} \right)^2 \sigma_{X_2}^2 + \dots + \left(\frac{\partial U}{\partial X_n} \right)^2 \sigma_{X_n}^2}$$

$$\sigma_U \leq \left| \frac{\partial U}{\partial X_1} \right| \sigma_{X_1} + \left| \frac{\partial U}{\partial X_2} \right| \sigma_{X_2} + \dots + \left| \frac{\partial U}{\partial X_n} \right| \sigma_{X_n}$$

5 Common Derivatives and Integrals

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\frac{d}{dx} c = 0$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\int \sin x dx = -\cos x + C$$

$$\frac{d}{dx} uv = u'v + uv'$$

$$\int uv' = uv - \int u'v$$

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\int \cos x dx = \sin x + C$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\int \sec^2 x dx = \tan x + C$$